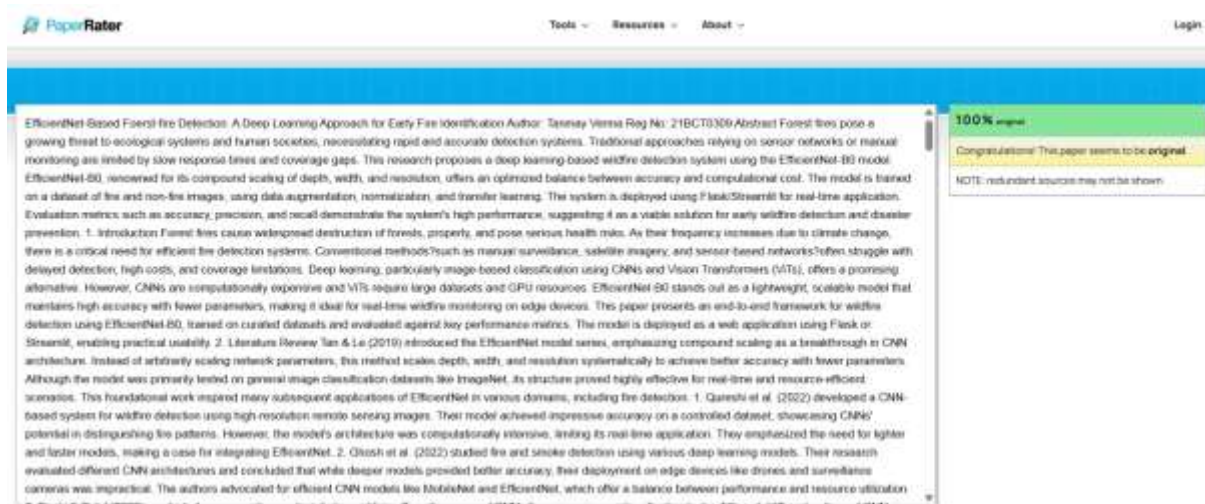


DESIGN OF SMART CITIES

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Plagiarism report



The screenshot shows a plagiarism report from PaperRater. The report is for a paper titled "EfficientNet-Based Forest fire Detection: A Deep Learning Approach for Early Fire Identification" by Tanmay Verma, Reg No: 21BCT0309. The report indicates that the paper is 100% original. The text of the report is as follows:

EfficientNet-Based Forest fire Detection: A Deep Learning Approach for Early Fire Identification Author: Tanmay Verma Reg No: 21BCT0309 Abstract Forest fires pose a growing threat to ecological systems and human societies, necessitating rapid and accurate detection systems. Traditional approaches relying on sensor networks or manual monitoring are limited by slow response times and coverage gaps. This research proposes a deep learning-based wildfire detection system using the EfficientNet-B0 model. EfficientNet-B0, renowned for its compound scaling of depth, width, and resolution, offers an optimized balance between accuracy and computational cost. The model is trained on a dataset of fire and non-fire images, using data augmentation, normalization, and transfer learning. The system is deployed using Flask/Streamlit for real-time application. Evaluation metrics such as accuracy, precision, and recall demonstrate the system's high performance, suggesting it as a viable solution for early wildfire detection and disaster prevention.

1. Introduction Forest fires cause widespread destruction of forests, property, and pose serious health risks. As their frequency increases due to climate change, there is a critical need for efficient fire detection systems. Conventional methods such as manual surveillance, satellite imagery, and sensor-based networks often struggle with delayed detection, high costs, and coverage limitations. Deep learning, particularly image-based classification using CNNs and Vision Transformers (ViTs), offers a promising alternative. However, CNNs are computationally expensive and ViTs require large datasets and GPU resources. EfficientNet-B0 stands out as a lightweight, scalable model that maintains high accuracy with fewer parameters, making it ideal for real-time wildfire monitoring on edge devices. This paper presents an end-to-end framework for wildfire detection using EfficientNet-B0, trained on curated datasets and evaluated against key performance metrics. The model is deployed as a web application using Flask or Streamlit, enabling practical usability.

2. Literature Review Tan & Le (2019) introduced the EfficientNet model series, emphasizing compound scaling as a breakthrough in CNN architecture. Instead of arbitrarily scaling network parameters, this method scales depth, width, and resolution systematically to achieve better accuracy with fewer parameters. Although the model was primarily tested on general image classification datasets like ImageNet, its structure proved highly effective for real-time and resource-efficient scenarios. This foundational work inspired many subsequent applications of EfficientNet in various domains, including fire detection.

1. Qureshi et al. (2022) developed a CNN-based system for wildfire detection using high-resolution remote sensing images. Their model achieved impressive accuracy on a controlled dataset, showcasing CNNs' potential in distinguishing fire patterns. However, the model's architecture was computationally intensive, limiting its real-time application. They emphasized the need for lighter and faster models, making a case for integrating EfficientNet.

2. Ghosh et al. (2022) studied fire and smoke detection using various deep learning models. Their research evaluated different CNN architectures and concluded that while deeper models provided better accuracy, their deployment on edge devices like drones and surveillance cameras was impractical. The authors advocated for efficient CNN models like MobileNet and EfficientNet, which offer a balance between performance and resource utilization.

100% original
Congratulations! This paper seems to be original.
NOTE: redundant sources may not be shown.